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Comparative analysis of OSINT tools, techniques, and legal aspects

Willi Lazarov^a ,* , Vojtech Moravec^a , Pavel Loutocky^b , Jakub Vostoupal^b ,
Zdenek Martinasek^a

^a Brno University of Technology, Antonínská 548/1, Brno 601 90, Czechia

^b Masaryk University, Žerotínovo nám. 617/9, Brno 601 77, Czechia

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ABSTRACT

Open Source Intelligence (OSINT) has emerged as a crucial practice in the digital era, driven by the rapid growth of information available on the internet and its expanding applications across multiple sectors. This study examines the role of OSINT as a vital tool in domains such as the indexed internet, social networks, the darknet, archives, and network devices. We present a comparative analysis of over 150 tools, services, and databases usable for OSINT. The analysis is based on publicly available information and highlights significant diversity in functionality, licensing, and accessibility, indicating that no single solution is capable of meeting all intelligence needs. The findings emphasize the necessity of combining multiple tools with manual methods to acquire accurate and reliable intelligence, while also highlighting persistent challenges, including limited integrations, licensing constraints, and the volatility of online sources. Beyond technical and methodological aspects, OSINT practice is shaped by complex legal and ethical considerations, as the public availability of data does not guarantee lawful use. This study provides a legal analysis of the General Data Protection Regulation (GDPR) and the Budapest Convention in relation to OSINT investigations. As compliance remains context-specific, the study underscores that the future of OSINT depends not only on technological advancement, but also on strong legal and ethical responsibility to mitigate risks of liability and reputational harm.

1. Introduction

Today, the internet is one of the largest sources of data. The amount of data is constantly growing, and the rate of growth is also steadily increasing by approximately 20%–30% each year. In 2023, the internet grew by approximately 147×2^{30} terabytes of data, followed by an increase of 181×2^{30} terabytes in the following year (Duarte, 2025). A significant portion of this data consists of open source content, such as social media posts, public documents, and multimedia files. It is also important to consider the deep web and darknet, which together account for more than 90% of the total data available on the World Wide Web (Dekker, 2025). This includes non-indexed parts of the internet, such as content accessible only after registering for a specific social network. Open data can represent valuable information, which has led to its collection and processing within Open Source Intelligence (OSINT).

OSINT as an intelligence discipline involves collecting, analyzing, and interpreting data to achieve a specific objective. In general, OSINT refers to information that has been identified, evaluated, refined, and

shared in alignment with a specific objective. The usage and OSINT applications are wide, from intelligence agencies, law enforcement, ethical hackers, journalists, to academic researchers (Hassan and Hijazi, 2018; Böhm and Lolagar, 2021). However, due to the easy availability of open source data, OSINT can also be used by threat actors. An example is the 2018 incident involving a server configuration error at the marketing company Exactis. This vulnerability caused the leak of the database with 340 million records — 230 million individuals and 110 million business customers (Heller, 2025). Other examples of the problematic use of OSINT include identity theft and its subsequent sale. This can be further facilitated by targeted phishing attacks or hybrid warfare, which combines cyber and physical attacks. In such cases, an actor can use geolocation techniques to track the location of a target and carry out an attack in physical space (Ford, 2022).

The legal landscape governing OSINT activities extends far beyond the notion of “public availability”. The erroneous presumption that publicly accessible information exists free from statutory constraints constitutes one of the most significant compliance risks facing

* Corresponding author.

E-mail addresses: lazarov@vut.cz (W. Lazarov), 219263@vut.cz (V. Moravec), loutocky@muni.cz (P. Loutocky), jakub.vostoupal@law.muni.cz (J. Vostoupal), martinasek@vut.cz (Z. Martinasek).

URLs: <https://www.vut.cz/en/people/willi-lazarov-221556> (W. Lazarov), <https://www.muni.cz/en/people/210290-pavel-loutocky> (P. Loutocky), <https://www.muni.cz/en/people/434784-jakub-vostoupal> (J. Vostoupal), <https://www.vut.cz/en/people/zdenek-martinasek-88846> (Z. Martinasek).

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practitioners today. OSINT operations must comply with multiple, overlapping legal frameworks that impose substantive limitations on data collection and processing. Therefore, in Section 5 of this paper, we delve into the relevant legal challenges and complexities, such as data protection regimes, particularly GDPR (European Union, 2016b); jurisdictional complexities from cross-border data flows; intellectual property violations from systematic data harvesting; contractual obligations through platform terms of service; and criminal liability including unauthorized access and identity fraud.

1.1. Key contributions

This paper addresses the current state of OSINT tools and techniques with reference to other intelligence disciplines. In our study, we present several key domains for open-source data and discuss legal aspects related to their collection and processing. The main contributions of our work are as follows:

- The comparative study includes over 150 tools and services divided into 7 categories, in which the identified tools are compared, described, and supplemented with a discussion of manual techniques.
- The paper is supported by an open-source dataset that we have published, enabling the community to further extend and maintain our work.
- We discuss specific aspects related to the selected OSINT scenarios and provide relevant legal recommendations on how to avoid associated risks.

To the best of our knowledge, limited work prior to this study has combined such a comprehensive analysis of OSINT tools with a detailed legal analysis and recommendations, which underscores our contribution to this domain.

1.2. Paper organization

The paper is divided into 6 sections. After the introduction in Section 1, the background of OSINT and a review of the related literature are presented in Section 2. The methodology of this study is defined in Section 3 and the key findings including all identified OSINT tools and services are presented in Section 4. This is followed by Section 5, which focuses on the legal aspects of OSINT. Lastly, Section 6 concludes the paper and outlines future directions.

2. Background and related work

Before presenting the results of an in-depth comparative analysis, this section provides background information and reviews the literature related to OSINT.

2.1. Open source intelligence

OSINT is one of the intelligence disciplines. It involves the collection, analysis, and interpretation of open source data, which includes sources such as social networks, online news, archives, and metadata. Although definitions of various intelligence disciplines may differ, they often overlap in certain areas and other intelligence disciplines, which include (Akhgar et al., 2016):

- **Human Intelligence (HUMINT)** — information from human sources.
- **Social Media Intelligence (SOCMINT)** — data from social media platforms.
- **Geospatial Intelligence (GEOINT)** — imagery and geospatial information.
- **Imagery Intelligence (IMINT)** — visual data from satellites, cameras, and other image sources.

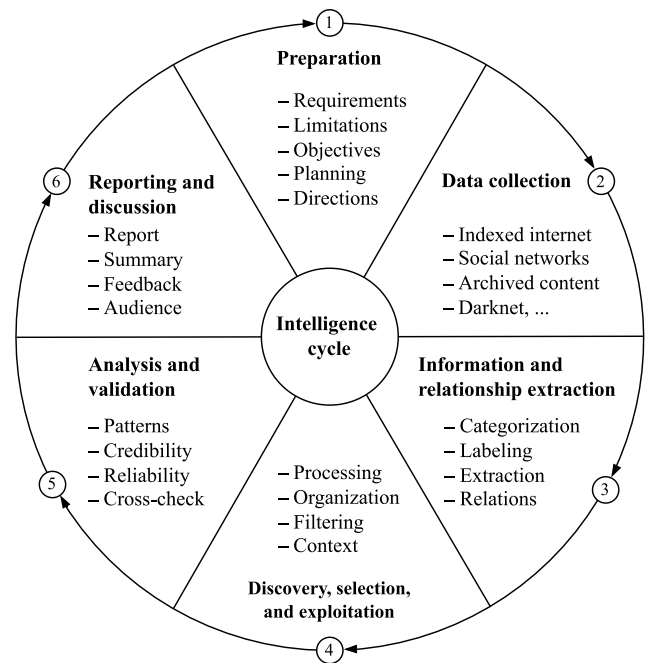


Fig. 1. Intelligence cycle.

- **Signals Intelligence (SIGINT)** — communication intelligence (COMINT) and electronic intelligence (ELINT) signals.
- **Measurement and Signature Intelligence (MASINT)** — distinctive characteristics (signatures) of specific objects or activities.

In OSINT, there are four basic components that are divided according to their intelligence value as follows:

- **Open source data (OSD)** — unprocessed (raw) data collected from various open sources such as websites, social networks, public databases, etc.
- **Open source information (OSIF)** — structured and interconnected data that express specific information.
- **Open source intelligence** — discovered, selected, and exploited data with intelligence value.
- **Validated OSINT (OSINT-V)** — verified information from reputable sources that increases the certainty and credibility of OSINT data.

The OSINT components are further placed into different phases of the intelligence cycle. We define the general intelligence cycle based on Akhgar et al. (2016) and extend it by incorporating Validated OSINT into the following consecutive steps: (i) preparation; (ii) data collection (OSD); (iii) information and relationship extraction (OSIF); (iv) discovery, selection, and exploitation (OSINT); (v) analysis and validation (OSINT-V); and (vi) reporting and discussion. The extended intelligence cycle and its key components are shown in Fig. 1.

2.2. Literature review

To present the state of the art in OSINT studies, we conducted an analysis of publications directly related to this topic. Specifically, we searched for review and research papers indexed in Google Scholar (Google LLC, 2025), Web of Science (Clarivate, 2025), and Scopus (Elsevier, 2025) using the terms “Open Source Intelligence” and “OSINT”. The scope of the literature review was limited to papers published between January 2015 and June 2025. We selected only publications explicitly focused on the OSINT domain as a whole, excluding single-purpose and limited applications. The other exclusion criteria were as follows:

Table 1
Comparison of OSINT-oriented papers.

Publication	Type	Content
Browne et al. (2024)	Review	AI-based OSINT research
Ghioni et al. (2024)	Review	Governance, ethical, legal, and social implications
Yadav et al. (2023)	Review	Application of OSINT in cybersecurity domain
Perez (2023)	Review	OSINT challenges and impact on human rights
Chaudhary and Bansal (2022)	Review	Countering terrorism using OSINT
Khan and Wallom (2022)	Research	Implementation of the proprietary OSINT system
Belghith et al. (2022)	Research	Study on social dynamics of OSINT investigations
Stodelov and Miloslavskaya (2022)	Research	OSINT tools and their integration
Yamin et al. (2022)	Research	Integrating OSINT with the Cyber Kill Chain
Hwang et al. (2022)	Review	Current state, security threats and requirements
Böhm and Lolagar (2021)	Research	OSINT investigations, legal and ethical aspects
Ungureanu (2021)	Research	Intelligence cycle and future of OSINT
Evangelista et al. (2021)	Review	Integrating OSINT with AI
Ju et al. (2020)	Review	Application of OSINT in military
Pastor-Galindo et al. (2020)	Research	Challenges and future trends of OSINT
Keliris et al. (2019)	Research	OSINT-based threat modeling in energy sector
Miller (2018)	Review	Defining OSINT and other intelligence disciplines
Gong et al. (2018)	Research	CTI data validation using OSINT feeds
Ziółkowska (2018)	Review	Military intelligence supported by OSINT
Sampson (2017)	Review	Criminal investigations using OSINT
Layton and Watters (2016)	Research	Automation of OSINT, privacy, and ethical aspects
Akhgar et al. (2016)	Review	Collective guide to OSINT investigations
Trottier (2015)	Research	OSINT investigations on social media

1. Publications not written in English.
2. Publications without full text availability.
3. Duplicate publications or preprints.

Based on these criteria and scope, we identified 23 relevant publications, which are listed in Table 1.

The OSINT literature covers the intersection of publicly available data, AI-driven methodologies, and their application in the cybersecurity, law enforcement, and intelligence domains. A prominent and widely discussed topic in recent publications, such as those by Browne et al. (2024), Ghioni et al. (2024), Yadav et al. (2023), and Evangelista et al. (2021), is the rapid integration of AI and machine learning into OSINT workflows to improve data collection, automation, and analysis. These papers emphasize that AI can significantly improve OSINT's efficiency and applicability, particularly in addressing complex challenges like cybercrime, national security, and threat detection.

From a different perspective, the review papers by Perez (2023) and Chaudhary and Bansal (2022) explore legal, operational and ethical challenges, especially those related to human rights and data protection. Additionally, Ungureanu (2021) highlights how technological advances have increased OSINT's relevance by enabling more powerful tools for open data exploitation. Similarly, Böhm and Lolagar (2021) discusses OSINT techniques, including digital privacy and legal concerns.

In addition to these aspects, Layton and Watters (2016) edited a collective book on automating OSINT, including methods, algorithms, and case studies. On a more general level, Hwang et al. (2022) presents the current state, challenges, and requirements of OSINT from a cybersecurity perspective. In contrast to earlier research, Miller (2018) discussed the reliance on open-source data in intelligence activities. More in-depth, Akhgar et al. (2016) edited a collection of 18 chapters presenting OSINT from theoretical perspectives to practical and legal aspects. From the social media domain, Trottier (2015) presents a study on how European law enforcement uses OSINT to monitor and analyze data from social networks.

Research by Stodelov and Miloslavskaya (2022) and Yamin et al. (2022) focuses on the integration of practical tools and operational use cases, with limited attention to theoretical or governance-related concerns. This contrast underscores the fragmented and multidisciplinary nature of OSINT research and domains such as cybersecurity (Yadav et al., 2023; Yamin et al., 2022; Gong et al., 2018), law enforcement (Sampson, 2017), critical infrastructure (Keliris et al., 2019), and military (Ju et al., 2020; Miller, 2018). Complementing these

views, Belghith et al. (2022) presents the collaborative dynamics and the use of tools for OSINT investigation. Within the practical use of OSINT, Khan and Wallom (2022) presents an implementation of the system for the management of open source data and information. However, in the review scope, we did not identify any study that examines OSINT tools across multiple domains in detail while providing a comprehensive comparative analysis.

Across the reviewed literature, several challenges emerge, such as the need for multilingual data support, resilience against misinformation and data misuse, ethical use of information, and improved integration of OSINT systems into real-world applications. However, notable gaps remain in standardized methodologies, cross-domain validation, and legislative frameworks for OSINT use, particularly as its adoption expands across government, military and civilian environment. The literature consistently calls for future research that balances technical innovation with social and ethical accountability.

3. Methodology

The comparative analysis of OSINT tools was based on data collected from publicly accessible sources during the period from January 2025 to August 2025. The primary sources used for data search included Google (2025), GitHub (2025), Medium (2025), and Nixintel (2025). The identification of OSINT tools and data sources was carried out through keyword-based searches, using various combinations of terms “OSINT“, “Open Source Intelligence“, and “data collection“, each paired with the descriptors “tool“, “application” and “system”. For more targeted results, these combinations were further combined with the names of specific social networks (e.g., Facebook, Instagram, TikTok) or other OSINT domains divided into the following categories:

- **Indexed internet** — search engines, clear web searchers, crawlers, and source codes.
- **Social networks** — tools, search engines, scrapers, downloaders, and official APIs.
- **Darknet** — onion site crawlers and monitoring tools.
- **Archived and cached content** — website snapshots, archived digital material, and cached pages.
- **Domains and IP addresses** — DNS and IP sources.
- **Network devices** — public devices, threat intelligence.
- **Multifunctional and specialized tools** — multiple domains, full stack tools, and AI-driven solutions.

All identified tools and data sources related to OSINT were subsequently labeled based on information available from official repositories, product websites, and technical documentation. The labeling process included the following attributes:

- **Category** — functionalities and techniques.
- **Open source** — distribution under open license.
- **Accessibility** — free or paid access.
- **API** — availability of an API.
- **Maintained** — official support (commercial solutions) or the latest commit within one year of the data collection period since January 2024 (open source).

In cases where reliable information could not be verified or where descriptions were ambiguous or inconsistent, the respective attribute label was omitted from the comparison table. Due to limited resources, the analysis was conducted exclusively using open sources, without performing any empirical benchmarking of the identified tools. The resulting comparative analysis of OSINT tools and sources, supplemented with relevant techniques and use cases, is presented in Section 4.

The presented study further investigated the legal aspects of OSINT. Our analysis is based primarily on the European Union law system and addresses issues of legality and privacy related to data collection from various open-source domains. To support this analysis, we prepared generalized scenarios representing different OSINT approaches, which serve as the basis for the legal assessment. The scenarios, including their sub-variants, are listed below and were selected based on the potential use of OSINT for investigations independent of law enforcement processes:

1. **Scenario 1.** Searching for information from a data breach published on the darknet:
 - (a) Personal data (name, e-mail, phone, etc.).
 - (b) Data of another person with their consent.
 - (c) Data of another person without their consent.
 - (d) Data of employees (e.g., by company email).
 - (e) Download the data breach and store it on disk for possible future use without searching in it.
 - (f) Download the data breach and upload it to a custom database connected to an online service.
 - (g) Keep the information on whether a person/data is present in the leak without persisting anything else.
2. **Scenario 2.** Monitoring online activity of children:
 - (a) Both parents request monitoring of their child's online activity using OSINT methods to prevent possible threats.
 - (b) The above (a), but the OSINT is carried out directly by one of the parents.
3. **Scenario 3.** Scanning web applications to deanonymize a person who committed a crime (following ethical hacking principles):
 - (a) The suspect who committed a crime owns a web application, and to identify this person, methods of active scanning of open ports, directories, downloading accessible repositories (e.g., .git), and other steps leading to identification of server misconfiguration are used (data collection only, no offensive operations).
 - (b) The above (a), but the web application of threat actor is hosted on a server shared with other users' applications.
 - (c) The above (a), but the web application is hosted in a public cloud.
4. **Scenario 4.** Verification of a person:
 - (a) A third party requests verification of a person in order to learn more about her/him and confirm information they have shared for the purpose of their own safety.

- (b) Verification of a person prior to acquaintance for reasons of personal safety.

5. **Scenario 5.** Gathering information that may serve in criminal and court proceedings:

- (a) The victim provides information gathered from publicly available sources (by a specialist) as evidence in a criminal complaint.
- (b) A person in a court case obtains information through a specialist that may help in the dispute.
- (c) The above (a, b), but the information is gathered by the victim themselves.

6. **Scenario 6.** Revealing the identity of the owner of a website, phone number, e-mail, IP address, username, or social media profile:

- (a) The findings may serve in criminal and court proceedings.
- (b) The findings may provide a person with information that can potentially prevent threats.

7. **Scenario 7.** Searching for people using only open sources:

- (a) Searching for a missing person.
- (b) Searching for a wanted person.
- (c) Searching for a debtor.
- (d) Searching for a family member.

8. **Scenario 8.** Competitive intelligence using only open sources:

- (a) Gathering information about competitors for competitive-ness.
- (b) Gathering information about competitors for future business planning.
- (c) Gathering information for business negotiations.
- (d) Gathering information for a public campaign.

9. **Scenario 9.** Gathering information from open sources:

- (a) Including personal data for personal use.
- (b) Excluding personal data for personal use.
- (c) Including personal data for commercial use.
- (d) Excluding personal data for commercial use.

10. **Scenario 10.** Creating a fake profile for OSINT:

- (a) The fake profile of a non-existent person is created, but it is passive and does not communicate with anyone.
- (b) The fake profile of a non-existent person is created, but it is active and communicates with other people (e.g., on Facebook).
- (c) The fake profile uses the photo of a different existing person.
- (d) The fake profile uses a photo of a non-existent person generated by AI.

4. Comparison of OSINT tools

We identified a total of 150 distinct solutions that are specifically designed for OSINT or can be used to collect open data and information. The comparison is organized in the following sections according to the categories defined in the comparative study methodology. All listed solutions and their sources are available on the OSINT Hub developed within this study. The data set is published as open source under the MIT license on GitHub ([Lazarov et al., 2025b](#)) and a public website (see [Appendix](#)).

4.1. Indexed internet

One of the most common ways to search for information and online resources (e.g., websites, e-books, source code, etc.) is through public search engines such as Google. These engines crawl websites, index and analyze their content, and then return the most relevant results based on the user's query. In this section, we compare well-known search engines and related tools that can be used to find and collect data from the indexed web, making them an essential component of the OSINT domain.

4.1.1. Comparative analysis

As shown in [Table 2](#), the majority of search engines provide advanced search functionality. In this regard, we list all supported search filters in the category column and the table legend. Given the vast and continuously increasing volume of data accessible on the internet, such features are not only useful but also essential for conducting efficient and precise searches. Another aspect examined in the comparative analysis was the availability of API (50%), although in some cases access is subject to specific restrictions or limitations (50%).

We also compared selected services from multiple perspectives, including the types of indexed data each tool is able to retrieve. These solutions are listed in [Table 3](#), where the most prevalent functionality is source code and content analysis (62.5%), which means the ability to retrieve historical content along with the source code of a given web page. Some services offer even more advanced capabilities, such as identifying the technologies employed in a specific domain (12.5%). Additional factors considered included the availability of the service and its open-source status. Our findings show that all selected services are either entirely free (37.5%) to use or provide at least a limited free version (62.5%). However, with regard to open-source availability, only one of the tools compared was released under an open-source license. The majority of tools (87.5%) also provide API access.

Although the compared search engines can provide results for most top-level domains (TLD), there are also local search engines (16.7%) specific to individual countries. In practice, the primary search is often performed using Google, Bing, or Yandex (GBY), complemented by a local engine such as [Seznam.cz \(2025\)](#) for Czech websites. A comprehensive list of available local search engines can be found on Search Engine Colossus ([Strome, 2025](#)), which aggregates country-specific search services from around the world.

4.1.2. Techniques and challenges

Despite the easy accessibility of indexed open sources, this domain faces several challenges, including frequent changes in search engine query syntax, result personalization and geo-bias, rate limiting, and the increasing use of anti-scraping mechanisms. To increase the accuracy of these tools, the key to searching over indexed data is using “dorking” on search engines. Dorking represents a parameterization based on filtering rules, such as `site` (matching domain), `inurl` (matching strings in URLs), and `intext` (matching strings in web content) in Google, which outputs only websites that meet these criteria. This can be further enhanced by the logical operators AND (logical conjunction) and OR (logical disjunction), or by specifying date intervals using `before` and `after`. Similarly, this approach can be applied to other search engines that support dorking or advanced (filtered) search.

4.2. Social networks

Social networks are online platforms that enable users to create profiles, share content, and interact with others. They contain an immense amount of user-driven data, including personal details, social connections, geolocation information, multimedia content, and behavioral patterns. For OSINT, these platforms are particularly important because they provide direct insight into individuals, groups, and organizations. The accessibility and informative value of data available on social

networks make them valuable sources for investigations, ranging from verifying identities, monitoring online activity, and uncovering hidden networks. At the same time, their dynamic nature and strict platform policies pose challenges for integration into OSINT workflows.

4.2.1. Comparative analysis

In this section, we compare the tools and services available on various social networking platforms that can be used as data sources for OSINT investigations. As shown in [Table 4](#), most of the selected tools are actively maintained (77.2%) and openly accessible (56.1%). The most common identified functionality is the retrieval of profile information (49.1%) such as first name, last name, nickname, and related data. From an open-source perspective, the proportion of open-source tools is relatively low (36.8%). However, most of the selected services remain freely accessible, although sometimes with limitations, as previously noted. In particular, the majority (77.2%) of these tools do not provide an API, which poses challenges for integration and limits opportunities for automation.

4.2.2. Techniques and challenges

In addition to the tools compared in [Table 4](#), SOCMINT, in combination with OSINT, faces challenges related to the strict policies of social platforms and the potential blocking of automated tools by anti-robot mechanisms such as the Completely Automated Public Turing test (CAPTCHA). As a result, despite the available data collection tools, manual OSINT techniques remain a crucial component of social network analysis. These techniques include profile and content analysis, pivoting across user connections and platforms, temporal analysis of posting behavior, and cross-verification of information using external sources. Their combination with automated tools and other intelligence disciplines helps mitigate platform limitations and improves the reliability and contextual understanding of collected social network data.

4.3. Darknet

Darknet¹ networks, such as Tor, the Invisible Internet Protocol (I2P), and Hyphernet, provide encrypted communication channels and host hidden services that are inaccessible through the conventional indexed internet ([Gehl, 2018](#)). In general, any regular service or website can be hosted on these networks, benefiting from added anonymity for both clients and providers. However, due to this enhanced privacy, darknet platforms play a central role in illicit marketplaces, underground forums, hacking services, and other concealed ecosystems ([Basheer and Alkhatib, 2021](#)). Unlike the surface web, darknet sites are intentionally unindexed, often transient, and protected by strong anonymity mechanisms, making them difficult to discover, monitor, and analyze. For OSINT purposes, this environment contains valuable sources of information for investigation (e.g., uncovering hidden communities or tracing digital artifacts) but also presents significant challenges, particularly in terms of accessibility ([Alquwayzani et al., 2023](#)).

4.3.1. Comparative analysis

We identified and compared 25 Tor tools and services with potential applicability for OSINT investigations in the darknet environment. As shown in [Table 5](#), the majority of these tools are actively maintained (60%), freely accessible (88%), and distributed mainly as open-source (76%). The most common functionality in these is the ability to identify onion sites related to a given keyword or set of keywords (44%). Another widely supported capability is metadata extraction (40%), which may include collecting email addresses, crawling

¹ In the literature and across various online sources, the terms “darknet” and “dark web” are often used interchangeably to describe this type of environment. Since these networks can support a wider range of services beyond websites, we consistently use the term “darknet” in this paper.

Table 2
Comparison of search engines.

Name	Category	API	Description
Google	SW, EW, AW, NW, L, R, T, D, F, IU, IT, !✓ ITX, P, RLT, N, SS, RU, U, IS	✓	Advanced query syntax for precise data retrieval
Yandex	SW, EW, AW, NW, IU, IT, L, R, D, T, P, !✓ SR, GR, IS, SS	✓	Comprehensive search with paid API access
Bing	SW, EW, AW, NW, F, E, IA, IB, IT, L, R, ✗ P, D, RF, PRF, SS, T, PREF, IP, RU, IS	✗	Rich search features; legacy API deprecation pending
Brave	E, F, IB, IT, IPG, L, R, D, SW, EW, AW, !✓ NW, SS, P, T	✓	Privacy-centric search with limited free API
DuckDuckGo	SW, EW, AW, NW, F, D, IT, IU, SS, P, T, ✓ IS, RU	✓	Privacy-first search with open API support
Yahoo	SW, EW, AW, NW, D, F, SS, R, L, N, P, – T, IS, RU	–	Legacy engine with basic advanced query tools
SearXNG	–	✓	Meta-search engine with customizable API
Baidu	–	–	Chinese-language search engine
Search Engine Colossus–	–	–	Directory of global search engine endpoints
I Search From	SW, EW, AW, NW, L, R, T, D, F, IU, IT, ✗ ITX, P, RLT, SS, RU, N, U, IS	✗	Geolocation manipulation for Google Search
Million Short	–	–	Search engine that skips popular sites.
BoardReader	Q, SW, EW, NW, L, R	✓	Forums and message boards search engine
Quantitative summary	Category: SW: 8x, EW: 8x, AW: 7x, NW: 8x, IU: 4x, IT: 6x, ITX: 2x, IB: 2x, IA: 1x, IPG: 1x, RLT: 2x, L: 7x, R: 7x, U: 2x, D: 7x, F: 6x, RU: 5x, T: 7x, P: 7x, SR: 1x, GR: 1x, IS: 6x, RF: 1x, PRF: 1x, SS: 7x, N: 3x, E: 2x, PREF: 1x, IP: 1x, Q: 1x, N/A: 4x; API: ✓: 3x, !✓: 3x, ✗: 2x, N/A: 4x.		

Q — Query, SW — Specific words, EW — Explicit words, AW — Any/All words, NW — Exclude words, IU — In URL, IT — In title, ITX — In text, IB — In body, IA — In Anchor, IPG — In Title or Body, RLT — Related pages, L — Language, R — Region, U — Last update, D — Domain, F — File type, RU — Rights of use, T — Time period, P — Pagination, SR — Sort by, GR — Grouping, IS — Image size, RF — RSS feed, PRF — Has feed, SS — SafeSearch, N — Results per page, E — Extension, PREF — Preferences, IP — Sites hosted by a specific IP address, !✓ — Limitation.

Table 3
Comparison of tools and databases for indexed internet.

Name	Category	Open source	Accessibility	API	Description
Site Explorer	S, SC, TA, ✗	✗	!✓	!✓	SEO audit, rank tracking, and site analysis tool
Common Crawl	CD	✓	✓	!✓	Open-source web crawl data archive
SeoReviewTool	S, BC	✗	✓	!✓	Provides backlinks and domain stats
NerdyData	SC, T	✗	!✓	!✓	Scrapes source code and detects used technologies
PublicWWW	SC	✗	!✓	!✓	Searches source code and keyword signatures
grep.app	SC	✗	!✓	!✓	Searches code, files, and paths on GitHub
Sourcegraph	SC	✗	!✓	!✓	Search for public code and files
Google Scholar	SL	✗	✓	✗	Scholarly literature
Quantitative summary	Category: S: 2x, SC: 5x, TA: 1x, CD: 1x, BC: 1x, T: 1x, SL: 1x; Open source: ✓: 1x, ✗: 7x; Accessibility: ✓: 3x, !✓: 5x; API: !✓: 7x, ✗: 1x.				

S — Site statistics, SC — Source code and content analysis, TA — Traffic analysis, CD — Crawl data, BC — Backlink analysis, T — Technology analysis, SL — Scholar literature, !✓ — Limited access.

websites, capturing screenshots, and related tasks. With regard to API availability, the findings are comparable to those in Section 4.2, that tools offering an API (12%) are the exception rather than the rule. This lack of API support complicates integration and makes automation challenging.

4.3.2. Techniques and challenges

The main challenges of OSINT in the darknet environment stem primarily from its anonymous and dynamic nature. Access to relevant information is complicated by technical barriers inherent in the functioning of the Tor and I2P networks. Unlike the indexed internet and registered domains that are publicly accessible, darknet sites remain fundamentally anonymous. Therefore, their discovery can be supported by sources such as onion site lists, addresses shared on discussion forums (e.g., Reddit), threat intelligence databases, and findings from intelligence activities.

4.4. Archived and cached content

Digital archives and caches provide a valuable source of information for OSINT, as they allow investigators to examine historical versions of websites or recover information that has been modified, deleted, or hidden from current sources. Such sources are particularly useful for browsing websites that no longer exist, verifying information, or

reconstructing digital footprints that would otherwise be inaccessible. However, not all sources are archived, which is caused by blocked or incomplete archiving, limited indexing of dynamic content, and intentional removal requests.

4.4.1. Comparative analysis

We identified and compared 7 services with archived and cached content that can be a valuable data source for OSINT investigations. In addition, we list open sources containing bibliographical content and leaked documents (e.g., classified materials). As shown in Table 6, the most common functionality is the ability to retrieve historical versions of websites (i.e., archived content) (57.1%). All of the compared services are also accessible without any usage restrictions. The majority of the tools do not offer an API (57.1%). Among those that do (42.9%), access is usually restricted (66.7%), which makes integration complicated. In the case of licensing, the majority of these tools and services are not distributed as open-source software (71.4%). Among all these sources, the Internet Archive plays an important role, as its database contains more than 946 billion archived web pages (Internet Archive, 2025).

4.5. Domains and IP addresses

Domain names and IP addresses represent one of the most fundamental layers of internet infrastructure and are therefore a critical part

Table 4

Comparison of tools and databases for social networks.

Name	Category	Social networks	Open source	Accessibility	API	Maintained	Description
Facebook Graph API	PI, PMC	Facebook	✗	!✓	!✓	✓	Official Meta API
Facebook Recover Lookup	PI	Facebook	✗	✓	✗	✓	Checks email-to-account match
Who Posted What	PI, PMC	Facebook	✗	✓	✗	✓	Searches Facebook posts by date
Facebook Graph Searcher	PMC, AUD, O	Facebook	✗	✓	✗	✓	Searches posts by time period
Graph.tips	PMC, PI, AUD, GEO, O	Facebook	✗	✓	✗	✓	Facebook search
LookupID	PI, G	Facebook	✗	✓	✗	✓	Extracts ID from a given URL
RandomTools	PI	Facebook	✗	✓	!✓	✓	Extracts ID from a given URL
Facebook Data Breach Checker	CN	Facebook	✗	✓	✗	✗	Checks if number was in 2019 FB breach
FDOWN	AUD	Facebook	✗	✓	✗	✓	Facebook video downloader
Tweepy	PMC, PI	X (Twitter)	✓	!✓	✗	✓	Twitter/X data collection via API
twarc2	PMC, PI	X (Twitter)	✓	!✓	✗	✗	Twitter/X data collection via API
Twint	PMC, PI	X (Twitter)	✓	✓	✗	✗	Twitter scraper without API
Twitter Advanced Search	O	X (Twitter)	–	–	–	–	Advanced Twitter/X search
Followerwonk.com	PI, S	X (Twitter)	✗	✓	✗	✓	Analyze/compare Twitter profiles
twitonomy.com	PI, S	X (Twitter)	✗	!✓	✗	✓	Twitter analytics
TWDown	AUD	X (Twitter)	✗	✓	✗	✓	Twitter/X video downloader
YouTube Data API	G, O	YouTube	✗	!✓	!✓	✓	Official YouTube API
yt-dlp	AUD, M	YouTube	✓	!✓	✗	✓	Video downloader
MW Metadata	S, M, GEO	YouTube	✓	!✓	✗	✓	Returns video metadata
TubePilot	M	YouTube	✗	!✓	✗	✓	Returns video metadata
YouTube GeoFinder	GEO	YouTube	✓	✓	–	✓	Finds geo-location of video
Anilyzer	FBF	YouTube, Vimeo	✗	✓	✗	✓	Frame-by-frame video viewer
Discord API	PI, PMC, O	Discord	✗	!✓	!✓	✓	Official Discord API
Discord.py	PI, PMC, +	Discord	✓	!✓	✗	✓	API wrapper
Discord History Tracker	PMC	Discord	✓	✓	✗	✓	Saves chat history
LinkedIn API	PI, G	LinkedIn	✗	!✓	!✓	✓	Official LinkedIn API
LinkedIn-scraper	PI, G	LinkedIn	✓	✓	✗	✓	LinkedIn scraper
TikTok-API	PI, M	TikTok	✗	!✓	!✓	✓	Official TikTok API
TikTok-scraper	PI, AUD, M, PMC	TikTok	✓	✓	✗	✗	Scrapes profile, video, comments
TikTok video downloader	AUD	TikTok	✗	✓	✗	✓	Downloads TikTok videos
TikTok date extractor	DT	TikTok	✓	✓	✗	–	Extracts video upload time
Exolyt	PI, T, AUD, S	TikTok	✗	!✓	✗	✓	TikTok trends and analytics
Trend Discovery	T	TikTok	✗	✓	✗	✓	Finds trending TikTok content
Instagram Scraper	M, PMC, GEO, PI	Instagram	✓	✓	✗	✗	Scraper for Instagram
Instaloader	PI, PMC, AUD	Instagram	✓	!✓	✗	✓	Instagram scraper
Famety	AUD	Instagram	✗	✓	✗	✓	Instagram media downloader
dumpor	PI, AUD, PMC	Instagram	✗	✓	✗	✓	IG viewer and data collector
Toolzu	PMC, AUD	Instagram	✗	✓	✗	✓	IG video/audio downloader
StorySaver	PMC	Instagram	✗	✓	✗	✓	IG story downloader
InstaHunt	PMC	Instagram	✗	!✓	✗	✓	Finds IG posts by location
InstaStoriesViewer	PMC	Instagram	✗	✓	✗	✓	Anonymous viewing of Instagram stories
Reddit API	PI, PMC	Reddit	✗	!✓	!✓	✓	Official Reddit API
PRAW	PMC	Reddit	✓	!✓	✗	✓	Reddit API wrapper
YARS	PI, PMC, AUD	Reddit	✓	✓	✗	✓	Reddit scraper
Pushshift API	PI, PMC, O	Reddit	✓	!✓	!✓	✗	Reddit comment/post API
Redective	PI, PMC, O	Reddit	✗	!✓	✗	–	Reddit profile info and stats
Reddit Comment Search	PMC	Reddit	✗	✓	✗	–	Searches comments by user/keyword
Pinterest API	PI, S	Pinterest	✗	!✓	!✓	✓	Official Pinterest API
VK API	PI, AUD, PMC, G	Vkontakte	✗	✓	✓	✓	Official VK API
OSINTvk	PI, AUD, G, GEO	Vkontakte	✓	✓	✗	✗	Intelligence gathering on VK
Python-telegram-bot	B	Telegram	✓	!✓	✗	✓	Telegram bot API wrapper
Pyrogram	B, PMC, AUD, G, +	Telegram	✓	!✓	✗	✓	Telegram API wrapper
Lyzem	C, G, B, PMC	Telegram	✗	✓	✗	✓	Searches channels/groups by keyword
TGStat	PMC	Telegram	✗	✓	!✓	✓	Searches posts/messages by keyword
PSE - Search TG Comments	PMC	Telegram	✗	✓	✗	✓	Searches Telegram comments
Telegram Nearby Map	GEO	Telegram	✓	!✓	✗	✗	Finds nearby Telegram users
Telegram Phone Number Checker	CN	Telegram	✓	!✓	✗	✗	Checks if number is on Telegram

Quantitative summary Category: PI: 28x, PMC: 29x, AUD: 16x, O: 7x, CN: 2x, G: 8x, GEO: 6x, S: 5x, T: 2x, FBF: 1x, M: 6x, DT: 1x, B: 3x, C: 1x, +: 2x; Social networks: Facebook: 9x, X (Twitter): 7x, YouTube: 6x, Vimeo: 1x, Discord: 3x, LinkedIn: 2x, TikTok: 6x, Instagram: 8x, Reddit: 6x, Pinterest: 1x, VKontakte: 2x, Telegram: 7x; Open source: ✓: 21x, ✗: 35x, N/A: 1x; Accessibility: ✓: 32x, !✓: 24x, N/A: 1x; API: !✓: 10x, ✗: 44x, ✓: 1x, N/A: 2x; Maintained: ✓: 44x, ✗: 9x, N/A: 4x.

PI — Profile information, G — Groups and communities, E — Email addresses, PMC — Posts, messages and comments, S — Statistical and analytical data, GEO — Geolocation data, CN — Cell phone numbers, AUD — Audio-visual content, B — Bot management, C — Channels, DT — Date and Time, T — Trends, M — Metadata, FBF — Frame-by-Frame video analysis, O — Other data types, + — and more, !✓ — Limited access.

Table 5

Comparison of tools and databases for darknet.

Name	Category	Open source	Accessibility	API	Maintained	Description
Katana	S, O	✓	✓	✗	✓	Google dorking scanner; supports Tor
OnionSearch	O	✓	✓	✗	✗	Keyword search across onion search engines
darkdump	O, SCDE	✓	✓	✗	✓	Scrapes onion sites by keyword
Ahmia	O	✓	✓	✗	✓	Search engine for Tor hidden services
Darkus	O	✓	✓	✗	✓	Keyword search across onion search engines
Onion Search Engine	O	✗	✓	!✓	✗	Keyword search across onion search engines
Tor2Web	O	✓	✓	✗	✓	Access onion sites via regular browsers
OnionLand Search	O	✗	✓	✗	✓	Access onion sites via regular browsers
IACA	O	✗	✓	✗	✓	Darknet and social media search platform
Hunchly	S, O, SCDE	✗	!✓	!✓	✓	Web capture tool for clearnet and darknet
OnionScan	IS	✓	✓	✗	✗	Scans onion sites for leaks and issues
docker-onion-nmap	IS	✓	✓	✗	✗	Nmap scanner for onion ports/services
ONIOFF	ST	✓	✓	✗	✗	Checks availability of onion links
OWASP TorBot	SCDE, ST	✓	✓	✗	✓	Crawls onion sites, extracts links and HTML
TorCrawl.py	SCDE	✓	✓	✗	✓	Crawls onion sites, extracts links and HTML
VigilantOnion	O, SCDE, CTI	✓	✓	✗	✗	Searches onion content by keyword
OnionIngestor	SCDE, CTI	✓	✓	✗	✗	Onion site scanner and monitor
darc	SCDE	✓	✓	✗	✓	Onion site crawler
midnight_sea	SCDE	✓	✓	✗	✓	Darknet marketplace investigation
pryingdeep	SCDE	✓	✓	✗	✗	Crawls onion for certs, crypto wallets, etc.
Lunar	CTI	✗	–	!✓	✓	Darknet monitoring for breaches and TTPs
deepdarkCLI	CTI	✓	✓	✗	✓	Threat collection from dark/deep web
spiderfoot	SCDE, CTI	✓	!✓	✗	✗	OSINT tool with darknet and Tor support
ExoneraTor	R	✓	✓	✗	✓	Shows when IP was last used as a Tor relay
Quantitative summary	Category: O: 11x, SCDE: 10x, S: 2x, IS: 2x, ST: 2x, CTI: 5x, R: 1x; Open source: ✓: 19x, ✗: 6x; Accessibility: ✓: 22x, !✓: 2x, N/A: 1x; API: ✗: 22x, !✓: 3x; Maintained: ✓: 15x, ✗: 9x, N/A: 1x.					

S — Sites related to keyword(s), O — Onion sites related to keyword(s), ST — Onion site status, CTI — Cyber Threat Intelligence, R — Tor relay history, SCDE — Scraping, Crawling and Advanced Data Extraction, IS — Infrastructure and Security Analysis, !✓ — Limited access.

Table 6

Comparison of tools and databases for archived and cached content.

Name	Category	Open source	Accessibility	API	Description
Internet Archive	A	✗	✓	!✓	View historical snapshots of websites
Archive.today	A	✗	✓	!✓	View archived versions of websites
CachedView.com	A	✗	✓	✗	Cached pages from Google and Archive.org
Anna's Archive	B	✓	✓	✓	Repository of books, papers, and magazines
Waybackpack	A	✓	✓	✗	View archived versions of websites
Cryptome	C	✗	✓	✗	Repository of leaked/secret government docs
WikiLeaks	C	✗	✓	✗	Repository of leaked/secret government docs
Quantitative summary	Category: A: 4x, B: 1x, C: 2x; Open source: ✓: 2x, ✗: 5x; Accessibility: ✓: 7x; API: ✓: 1x, !✓: 2x, ✗: 4x.				

A — Archived content, B — Bibliographical content, C — Leaked and sensitive documents and information, !✓ — Limited access.

of OSINT investigations. Along with certificates, they provide direct insight into the ownership, hosting, and configuration of online resources (e.g., web servers), enabling investigators to find relationships between entities, track digital footprints, and identify potential vulnerabilities. By analyzing publicly available records such as WHOIS data, DNS resolutions, and public IP addresses, investigators can map connections between websites, organizations, and hosting providers. Unlike content-based analysis, which focuses on what is published online, examination of domain and IP data reveals the technical backbone that supports those services, often exposing hidden linkages or previously unrecognized infrastructure components.

4.5.1. Comparative analysis

In Table 7, we compare a range of services that can track and provide information about domains, certificates, and IP addresses. Most of these sources provide details about domain registration, hosting, SSL/TLS certificates, and other related data. Some of the compared tools are more versatile, offering additional information about a given target or supporting the combination of multiple search criteria (45.5%). While most of the tools reviewed are not open source (90.9%), all of them are freely accessible. However, this often comes with certain limitations, such as a limited number of daily requests. The same applies to their APIs, which are commonly available (81.8%) but typically subject to usage restrictions.

4.5.2. Techniques and challenges

Despite the availability of automated tools and services, OSINT focused on domains and IP addresses poses several challenges. These include incomplete or obfuscated WHOIS records due to privacy protection services, historical domain data retention changes, rate limiting, and inconsistencies across data providers. Additionally, shared hosting environments and cloud infrastructure can complicate attribution by associating multiple unrelated domains with a single IP address. To address these challenges, manual techniques can be used to increase the accuracy of OSINT investigations such as correlating WHOIS records with historical snapshots, finding links between DNS records to identify domain relationships, and tracking changes over time to detect domain migrations or ownership transfers.

4.6. Network devices

Network device indexing services provide large-scale visibility into the state of the internet by cataloging connected devices and their configurations. These platforms scan and record technical details, including IP addresses, open ports, service banners, software versions, and sometimes even geolocation data (Bennett et al., 2021). For OSINT, such data can reveal the digital exposure of organizations, detect misconfigured or vulnerable systems, and uncover the online presence of entities that may not be visible through conventional search engines.

Table 7
Comparison of tools for domains, certificates and IP addresses.

Name	Category	Open source	Accessibility	API	Maintained	Description
DomainTools	D, IP	✗	!✓	!✓	✓	Domain lookup and monitoring
WhoisFreaks	D	✗	!✓	!✓	✓	Aggregated WHOIS data with APIs
WHOXY	D	✗	✓	!✓	✓	Reverse WHOIS and domain research
Whoisology	D	✗	!✓	✗	✓	WHOIS records and connections
IPHunter	IP	✗	!✓	!✓	✓	Tracks IP addresses
crt.sh	C	—	✓	✗	✓	Certificate Transparency search engine
SecurityTrails	D, IP, C	✗	!✓	!✓	✓	Domain, IP intelligence platform
URLScan	D, IP, O	✗	!✓	!✓	✓	URL analysis and threat intelligence
MXToolBox	D, IP, O	✗	!✓	!✓	✓	Email, DNS, and blacklist lookup tool
ViewDNS.info	D, IP, O	✗	✓	!✓	✓	Domain and IP intelligence lookup
DNSlytics	IP, D, O	✗	!✓	!✓	✓	Domain, IP, and DNS intelligence
Quantitative summary	Category: D: 9x, IP: 7x, C: 2x, O: 4x; Open source: ✗: 10x, N/A: 1x; Accessibility: ✓: 3x, !✓: 8x; API: !✓: 9x, ✗: 2x; Maintained: ✓: 11x.					

D — Domain, IP — IP addresses, C — Certificates, O — Other, !✓ — Limited access.

Unlike standard web content, information about network devices is often more persistent and less structured, offering a raw view into the operational footprint that can be further processed.

4.6.1. Comparative analysis

We identified and compared services that index and cache information about devices connected to the internet, including details such as IP addresses, open ports, services running on those ports, service versions, and related metadata. As shown in Table 8, none of the selected services is distributed under an open-source license. Furthermore, the use of most of these platforms is subject to restrictions unless a commercial license is purchased (80%). Such limitations affect both access to the platform itself, such as the number of searches permitted and the scope of the results returned, including the number of services listed in a report, the vulnerabilities displayed per service, and other related outputs.

4.6.2. Manual techniques and challenges

OSINT focusing on network devices faces several challenges related to data accuracy and accessibility, as indexing services rely on periodic active scanning, which may result in outdated or incomplete information due to the dynamic nature of the global internet. Changes such as IP address reallocation, network reconfiguration, and service redeployment can significantly affect the timeliness and reliability of scanning results. To mitigate these limitations, we recommend to aggregate data from multiple sources in order to increase coverage, improve data completeness, and reduce bias introduced by individual scanning services. Correlating aggregated results using IP addresses, service banners, and device metadata supports validation of findings and uncovering relationships between devices that may not be apparent when relying on a single source.

4.7. Multifunctional and specialized tools

While tools and services focused explicitly on specific open sources provide valuable insights into individual online domains, there are also specialized tools (e.g., email lookups, threat intelligence, etc.) and OSINT toolsets capable of aggregating data from multiple sources. Multifunctional tools address this need by offering all-in-one platforms that extend beyond a single domain as in previous comparisons. These solutions integrate diverse capabilities such as account discovery, social media monitoring, and metadata extraction, enabling OSINT investigators to conduct more comprehensive inquiries within a unified environment (Stodelov and Miloslavskaya, 2022). However, there are limitations such as the lack of specialized utilities, the absence of certain open sources, or the reliance on outdated datasets when integrations are not regularly maintained.² The breadth of collected data

can also generate noise, requiring careful validation and analysis (Weir, 2016).

4.7.1. Comparative analysis

The comparison in Table 9 involves all-in-one and other types of tools focused on OSINT and related intelligence disciplines. Regarding the data that these tools can obtain, the most common is profile information (30%), which includes first name, last name, username, email, and other related data. For example, Sherlock can determine whether an account with a given username exists on more than 400 services, including Facebook, Instagram, and GitHub. Another example is Maltego, an all-in-one platform for complex OSINT investigations that offers a wide range of capabilities. Additional functionality identified among the selected tools is retrieving information from cryptocurrency wallets, collecting posts and comments from social media platforms (10%), extracting geolocation data (15%), and more. From a licensing perspective, most of these tools are not open-source (80%). However, at least limited free access is provided in most cases (80%). Similarly, this applies to restricted (paid) API access. Lastly, all the OSINT tools reviewed are actively maintained. Given the growing relevance of artificial intelligence, we also discuss platforms integrating AI into OSINT in Section 4.7.2.

4.7.2. Integration of AI into OSINT

The emergence of artificial intelligence has introduced new capabilities into OSINT investigations, particularly in processing large volumes of data and automating analytical workflows. AI-enabled tools extend traditional OSINT methods by applying techniques such as natural language processing, anomaly detection, and automated entity extraction. These systems support investigators by aggregating information from diverse sources including social media, threat intelligence platforms, darknet, and more, and transforming it into structured intelligence.

Compared to conventional tools focused on specific data types, AI-driven solutions offer broader analytical depth but also rely heavily on continuous model updates to maintain accuracy in rapidly changing information environments. The comparison in Table 9 includes several representative tools that integrate AI to enhance OSINT investigations. Most of them operate across multiple domains (83.3%), providing capabilities such as real-time risk detection, automated data collection, and AI-assisted investigative analysis. For example, Taranis AI aggregates and evaluates large information streams, while platforms such as Dataminr and Babel X serve as risk intelligence systems capable of identifying emerging events through machine learning models. Other solutions, such as Paliscope and Maltego, integrate AI components within a broader scope, supporting pattern recognition and assisted correlation.

As shown in Table 9, these tools are predominantly proprietary, often provide restricted access, and typically offer APIs for integration.

² The need for sustainability is crucial for OSINT tools, as open sources are continuously created, updated, and even destroyed over time.

Table 8

Comparison of tools and databases for network devices.

Name	Category	Open source	Accessibility	API	Services	CVE	Description
Shodan	IP, H, D, ISP, GEO, OS, T, P, SV, C, V	✗	!✓	!✓	✓	✓	Search engine for internet-connected devices
Censys	IP, OS, P, SV, GEO, C, D, V	✗	!✓	!✓	✓	✓	Real-time internet intelligence and threat insights
ONYPHE	IP, H, D, T, V, +	✗	!✓	!✓	✓	✓	Cyber threat intelligence and attack surface data
ZoomEye	IP, P, SV, V, GEO, C	✗	!✓	!✓	✓	✓	Internet-connected devices and vulnerabilities
Coalition Control	–	✗	–	✓	–	–	Cyber insurance with security and threat monitoring
Fofa	IP, H, D, P, SV, GEO, V, OS, T	✗	!✓	!✓	✓	✓	Chinese search engine for internet-connected devices
FullHunt	IP, D, GEO, P, SV, V	✗	!✓	!✓	✓	✓	Monitoring external digital assets
Netlas	IP, D, GEO, P, SV, V, C	✓	!✓	!✓	✓	✓	Internet assets and attack surface analysis
Odin	IP, GEO, ISP, P, V, OS	✗	!✓	!✓	✓	!✓	Internet-scanning search engine
NERD	IP, GEO, H, +	✓	✓	!✓	✗	✗	Open-source threat intelligence for network entities
Quantitative summary	Category: IP: 9x, H: 4x, D: 6x, ISP: 2x, GEO: 8x, OS: 4x, T: 3x, P: 7x, SV: 6x, C: 4x, V: 8x, +: 2x, N/A: 1x; Open source: ✓: 2x, ✗: 8x; Accessibility: ✓: 1x, !✓: 8x, N/A: 1x; API: !✓: 9x, ✓: 1x; Services: ✓: 8x, ✗: 1x, N/A: 1x; CVE: ✓: 7x, !✓: 1x, ✗: 1x, N/A: 1x;						

IP — IP address, H — Hostname, D — Domain, GEO — Geolocation data, T — Technologies used, OS — Operating system, P — Open ports, SV — Services, V — Vulnerabilities, ISP — Internet service provider, C — Certificate, + — and more, !✓ — Limited access.

Table 9

Comparison of multifunctional and specialized tools.

Name	Category	Open source	Accessibility	API	Maintained	Description
Sherlock	PI	✓	✓	!✓	✓	Finds accounts by username
Hunter.io	E	✗	!✓	!✓	✓	Email lookup tool
Facepager	PI, PMC, O	✓	✓	✗	✓	Collects public data from platforms
Social Searcher	PMC	✗	✓	–	✓	Social media search engine
Maltego	FS, AI	✗	!✓	!✓	✓	Complex AI-driven tool
Social Blade	S, GEO	✗	✓	!✓	✓	Platform stats and geolocation
PhantomBuster	PI	✗	!✓	!✓	✓	Data extractor for many platforms
osint.rocks	PI, E, O, CN	✗	!✓	✗	✓	Curated directory of OSINT tools.
IntelTechniques	PI, E, O, GEO	✗	✓	✗	✓	Browser-based OSINT search aggregator
Deepware	DD	✗	!✓	!✓	✓	Deepfake detection
EmailRep	E	✗	!✓	!✓	✓	Email reputation
WhatsMyName	PI	✓	✓	✗	✓	Finds accounts by username
DeHashed	E, O, +	✗	!✓	!✓	✓	Breach search and monitoring
VirusTotal	O	✗	!✓	!✓	✓	Threat intelligence platform
Synapsint	E, CN, GEO, O	✗	–	!✓	✓	Comprehensive OSINT platform
Taranis AI	AI	✓	✓	–	✓	AI-powered aggregation and analysis.
Babel X	FS, AI	✗	✗	–	✓	Risk Intelligence Platform
Dataminr	FS, AI	✗	✗	!✓	✓	Risk Intelligence Platform
Paliscope	FS, AI	✗	!✓	!✓	✓	Digital investigation and data analysis
DarkTrace	FS, AI	✗	✗	–	✓	Digital investigation and data analysis
Quantitative metrics	Category: PI: 6x, PMC: 2x, O: 6x, E: 6x, FS: 5x, S: 1x, GEO: 3x, CN: 2x, DD: 1x, +: 1x, AS: 1x, AI: 6x; Open source: ✓: 4x, ✗: 16x; Accessibility: ✓: 7x, !✓: 9x, ✗: 3x, N/A: 1x; API: !✓: 12x, ✗: 4x, N/A: 4x; Maintained: ✓: 20x.					

PI — Profile information, E — Email address, CN — Cell phone number, PMC — Posts, messages and comments, FS — Full stack (Social media, dark web, breaches and more), S — Statistical and analytical data, GEO — Geolocation data, DD — Deep Fake detection, AI — Artificial intelligence integration, O — Other data types, + — and more, !✓ — Limited access.

All reviewed solutions are actively maintained, reflecting the necessity of regular updates for sustaining AI performance and data relevance. However, the use of AI in OSINT also introduces notable limitations. AI-driven systems, such as Large Language Models (LLMs), may propagate biases present in training data, misinterpret context, or generate plausible but incorrect outputs (Kamath et al., 2024). These limitations underscore the critical need for systematic OSINT validation. Therefore, AI-generated findings should be corroborated through independent sources, cross-checked with traditional analytical methods, and reviewed by human analysts to ensure accuracy, reliability, and ethical use.

4.8. Use cases

In this section, we present several use cases as examples of how OSINT can be utilized in real-world scenarios, including references to tools listed in previous comparisons. OSINT is a wide domain and one approach cannot be generalized due to the high number of data sources, their variability, and dynamic nature. Therefore, the process of selecting OSINT tools and techniques depends on the objectives and targeted open-source domains.

In breach investigations, OSINT techniques help analysts quickly assess the scope of a compromise, identify leaked information, and

uncover potential threats. Tools such as DeHashed support the verification of exposed credentials and EmailRep provides intelligence on email reputation. For OSINT related to anonymization networks, specialized tools focused on the darknet (e.g., ExoneraTor) can confirm whether an IP address was part of the Tor network at relevant times. When investigating activity on the Tor network, specialized tools (e.g., Ahmia, OnionSearch, VigilantOnion) and darknet crawlers (e.g., darkdump, TorCrawl.py, OnionScan) can help identify leaked data, dark marketplaces, or conversations related to investigations.

Another significant area where OSINT delivers high operational value is threat intelligence, as it enables the systematic collection and analysis of open-source indicators related to emerging or ongoing cyber threats. OSINT tools can support the identification of exposed services (e.g., Shodan, Censys), mapping of adversary infrastructure and domain history (e.g., SecurityTrails, ViewDNS, crt.sh), correlation of entities and contextual enrichment (e.g., Maltego), and monitoring of threat-actor activity on public or semi-open platforms (e.g., Social Searcher). Threat intelligence platforms (e.g., VirusTotal) further enhance this workflow by enabling rapid enrichment of indicators of compromise and pivoting across related samples, helping analysts detect malicious files, URLs, IP addresses, and domains.

Another possible use case of OSINT is the verification of third-party vendors before establishing or maintaining business relationships. In this case, OSINT provides independent verification of a supplier's reputation, digital footprint, and security posture. Tools in domain, IP, and certificate intelligence categories (e.g., SecurityTrails, ViewDNS, crt.sh, URLScan, MXToolBox) support mapping the supplier's infrastructure and highlighting misconfigurations or risky exposures. Solutions like Hunter.io can validate corporate contact information, while dorking on search engines can identify related sources and information for further analysis. Additionally, platforms such as Sherlock or WhatsMyName allow light-touch checks of public user profiles associated with key individuals.

4.9. Summary

The comparative analysis covered over 150 distinct tools, databases, and data collection services that can be used for OSINT in multiple domains, including the indexed internet and search engines, social media networks, darknet environments, domains, certificates and IP addresses, archives, and network devices. Although the analysis did not include empirical benchmarking of the identified tools, our findings indicate that while the number of available OSINT solutions is substantial, they vary significantly in both functionality and accessibility.

One of the identified attributes across categories is the prevalence of free-to-use services, many of which offer limited access, while open-source availability remains underrepresented outside specialized areas such as darknet tools. API support is also inconsistent. While common among indexed internet and multifunctional platforms, it is often missing in tools for social networks and darknet environments, limiting automation and large-scale integration.

In terms of functionality, search engines and archived content providers primarily support structured retrieval of web sources, while social media and darknet tools enable OSINT investigation of dynamic, user-driven information. Domain, certificate, IP address, and network device platforms, on the other hand, provide technical insight into the internet infrastructure, exposing relationships, vulnerabilities, and digital footprints beyond surface-level content. Multifunctional tools attempt to bridge these domains by aggregating data sources into unified environments, but they sometimes trade specialization for breadth.

Despite the breadth and diversity of available OSINT tools, the comparative analysis highlights several systemic challenges that limit their practical use, including fragmented ecosystems, inconsistent or absent API support, restrictive access models, and a low proportion of open-source solutions. Fully resolving these issues is unlikely, as many

constraints stem from inconsistent structure, platform policies, commercial interests, and legal considerations. However, their impact can be mitigated by hybrid workflows that balance automated collection with manual techniques. In particular, the adoption of common data formats, clearer documentation of tool limitations, and greater support for partial automation can improve interoperability, even where full integration is not feasible.

Overall, the OSINT landscape is characterized by a large number of different solutions, high variability, and dependence on various investigative methods. Effective OSINT workflows therefore benefit from a combination of specialized tools, multifunctional platforms, open databases, and manual techniques, enabling investigators to adapt to diverse sources, access restrictions, and investigative goals. Furthermore, despite the use of open sources, OSINT techniques and data collection remain subject to legal constraints that may limit their use, as discussed in Section 5.

5. Legal aspects

Although the analyzed OSINT tools operate mostly with publicly accessible data, they do not exist in a legal vacuum. The mere fact that information is openly available does not automatically mean that it can be freely collected, processed, and reused without restrictions. OSINT practitioners who are interested in using the tools analyzed in Section 4 must therefore navigate a complex regulatory landscape in which different legal regimes overlap: data protection law, intellectual property law, contractual terms of platforms, and criminal law.

Although criminal law remains largely unharmonized, many relevant aspects are regulated within the Budapest Convention on Cybercrime and its Protocols (Council of Europe, 2025). Furthermore, it also applies in situations where the OSINT would be used by state agencies and provides an international legal framework for data collection and investigation in connection with criminal activity. It is therefore a highly relevant piece of legislation.

In this section, we focus on the European Union (EU) legal framework. The EU was chosen because its regulatory environment — particularly the General Data Protection Regulation (GDPR) (European Union, 2016b), the Law Enforcement Directive (LED) (European Union, 2016a) and the Budapest Convention³ (Council of Europe, 2025) — represent one of the most comprehensive and influential approaches to data protection, law enforcement processing, and cybercrime regulations worldwide. Furthermore, EU rules have extraterritorial reach, applying whenever personal data of EU citizens is processed, regardless of where the practitioner is located.

However, it should be emphasized that the use of OSINT can lead to a wide range of situations with distinct legal implications, from relatively unproblematic searches in open government databases to highly sensitive practices such as the processing of breach data or the creation of fake online profiles. Unfortunately, a full-scale legal analysis of all relevant legislation would go far beyond the scope of this article. Instead, we provide an overview of key regulatory areas and their practical consequences, complemented by model scenarios that illustrate how legality can vary depending on context. The goal is not exhaustive coverage, but to highlight the most pressing risks and recurring patterns OSINT practitioners must be aware of. Based on these reasons, apart from delving into particular legal areas, we present model scenarios within Table 10.⁴

³ While the Budapest Convention is not a product of the EU legislature procedure, it binds the Member States nevertheless, which means that the EU presents a uniquely transparent legal environment in this matter.

⁴ Considering the number of analyzed tools, it would be impractical to provide an all-encompassing, national-specific legal analysis.

5.1. Privacy protection

GDPR may constitute (and in many situations will indeed constitute) the primary legal framework governing personal data processing in OSINT activities within the EU. Its violations can lead to severe administrative fines, investigations by supervisory authorities, and orders to erase unlawfully processed data, which many OSINT practitioners may prefer to avoid.

The question of whether GDPR applies to OSINT activities or not is found in Article 2 par. 2 letter (c) and (d), which exclude processing of personal data by a natural person in the course of a purely personal or household activity; and by competent authorities for the purposes of the prevention, investigation, detection, or prosecution of criminal offenses or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security. The main limitation of Article 2 par. 2 letter (c) is that any activity related to professional or commercial activities cannot be considered a personal or household activity. This provision must be interpreted restrictively (Nulíček et al., 2018). According to the (CJEU, 2003) ruling in Case C-101/01 Lindqvist, even the mere publication of information/personal data about a natural person on a website does not fall under the exception for personal or household activities.

In cases where GDPR applies, OSINT researchers become controllers of the personal data gathered during the research and therefore need to fulfill all the obligations prescribed by GDPR, among others, the need to specify the proper legal basis for collecting personal data. OSINT investigations most often rely on the legal basis of legitimate interest (Article 6 par. 1 letter f) GDPR) (OSINT Central, 2023). However, this requires a balancing test between the interest pursued and the rights of the data subject. Failure to conduct or document this balancing test exposes the practitioner to liability.

The OSINT activities must then comply with core principles, including lawfulness, fairness, transparency, purpose limitation, accuracy, storage limitation, and accountability (Article 5 GDPR). The data minimization principle is particularly relevant, requiring collection of only data necessary for the specific purpose. That may be especially challenging in OSINT contexts, where data collections are often broad and exploratory.

When personal data are not obtained directly from the data subject — as is typical in OSINT — controllers must provide information to data subjects within one month (Article 14 GDPR), unless exemptions under Article 23 apply. These exemptions include prevention, investigation, detection or prosecution of criminal offenses, and protection of data subjects or the rights of others.

Specifically, it is therefore necessary to examine in the relevant Member State whether, in the case of national security, investigation and prevention of criminal offenses, and other conditions mentioned, it was not possible to refrain from informing the subject of such processing.⁵ When using such an exception (if it exists), it is particularly necessary to introduce enhanced safeguards, such as specifying the purpose of processing, determining the period of storage and handling of data, increasing the level of security, and striving to minimize the handling of personal data.

Special risks arise when dealing with children's data (consent requirements for children under 16 and other protection enhancements within Article 8) or special categories of personal data (political opinions, religious beliefs, health-related information). Inappropriate handling may quickly shift an otherwise lawful OSINT activity into non-compliance.

In situations where OSINT is employed by law enforcement, GDPR would not apply. Instead, LED specifically regulates the processing of

personal data by law enforcement authorities for criminal investigations. It provides rules and safeguards that balance effective crime prevention with individuals' privacy rights, offering exemptions from some GDPR obligations when transparency might jeopardize investigations. The LED ensures that data collection and processing methods used by law enforcement, including OSINT, comply with the legal standards necessary for admissible evidence and cross-border cooperation, making it indispensable for lawful and accountable OSINT operations in criminal justice contexts.

Finally, in situations where OSINT researchers would satisfy the requirements of the 'purely personal or household activity' exemption, the GDPR would also not apply. Instead, it is necessary to respect the general principle of privacy protection, which is incorporated at the EU level by the European Union Charter of Fundamental Rights (European Parliament, 2000). Article 7 protects the respect for private and family life, home, and communications. This protection extends beyond physical spaces to include informational privacy and digital communications, making it directly applicable to modern OSINT operations. Therefore, in cases where the data being processed are not personal data, but data that infringe on privacy, it is also necessary to comply with this basic limiting fundamental right, which is, however, generally limited, particularly in relation to the activities of law enforcement bodies.

5.2. Intellectual property and database rights

Beyond data protection, OSINT practitioners face risks under copyright law and the sui generis database right, an often overlooked part of the legal environment. Systematic scraping or archiving of content may infringe the rights of authors or database producers (Directives 2001/29/EC⁶ and 96/9/EC⁷). However, recognizing protected content can be quite challenging, even for lawyers. Usually, databases of a regular, technical nature without much of a specific creative organization should not be protected under copyright laws. Analogously, OSINT researchers can scrape databases protected by sui generis protection as long as they do not scrape a substantial part of them (qualitatively or quantitatively) and they do not reuse them (e.g., by publishing them).

Civil law consequences include injunctions, damage, and destruction of infringing copies. Importantly, intellectual property infringements can occur even if the data is publicly accessible, highlighting that "publicly available" does not equal "free to use".

Fortunately, the still rather new DSM Directive⁸ provides in its Article 4 an exception for text and data mining even for databases that are granted copyright or sui generis protection (however, it is still not possible to publish it and claim authorship). However, the setting of exceptions is specific to each Member State, and it is therefore necessary to respect the specific legislative settings in the relevant Member State where the relevant data are located (or which legal regime they follow).

5.3. Platform terms of service

Most OSINT activities are conducted on or against large online platforms (social networks, search engines, archives). Each platform imposes its own terms of service, which often prohibit automated scraping (or scraping in general, e.g., LinkedIn T&S), creating fake profiles, or impersonation. Violations may lead to:

⁶ Directive 2001/29/EC of the European Parliament and of the Council of 22 May 2001 on the harmonization of certain aspects of copyright and related rights in the information society (European Union, 2001).

⁷ Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases (European Union, 1996).

⁸ Directive (EU) 2019/790 of the European Parliament and of the Council of 17 April 2019 on copyright and related rights in the Digital Single Market and amending Directives 96/9/EC and 2001/29/EC (European Union, 2019).

⁵ See the specific guidelines on implementation by European Data Protection Board (2020).

- Account suspension or deletion.
- Loss of access to collected data.
- Civil actions in cases of large-scale or commercial breaches of contract.

Furthermore, cases such as *hiQ Labs v. LinkedIn* demonstrate that platform operators actively litigate against OSINT-type data collection. Even when courts do not always side with platforms, practitioners must be aware of potential litigation costs and reputational risks (Böhm and Lolagar, 2021). However, it is necessary to emphasize that in certain cases, there might be a reason to outweigh the protection safeguarded by said terms and conditions, such as a criminal investigation in cases of missing persons. In these cases, even the tools otherwise perceived as potentially fundamentally illegal or capable of breaching terms and conditions (e.g., LinkedIn scraper) might be used legally and legitimately, because the benefits outweigh the social harm.

5.4. Criminal law exposure

Certain OSINT practices may trigger criminal liability, even when the practitioner did not intend to act unlawfully. Relevant offenses include unauthorized access to information systems, identity fraud or impersonation, stalking or harassment, and handling unlawfully obtained data. The definitions and prevention of these offenses are provided in the following subsections.

5.4.1. Unauthorized access to information systems

- Definition: Any unauthorized access that goes beyond what is publicly visible may be qualified as hacking (e.g., active scanning).
- Prevention⁹: limit investigations to data that are genuinely public; avoid intrusion techniques (port scanning, brute-forcing, bypassing access restrictions). Use only official APIs or openly accessible sources.

5.4.2. Identity fraud or impersonation

- Definition: Creating fake profiles or misusing the photos of real persons can be considered impersonation or fraud.
- Prevention: Avoid using real identities or photos without consent; if dummy accounts are necessary, use AI-generated avatars with clear internal documentation and avoid active deception (communication, manipulation).¹⁰

5.4.3. Stalking or harassment

- Definition: Systematic monitoring of individuals, especially in a personal context, may amount to harassment.
- Prevention: Restrict monitoring to contexts where there is a clear and lawful purpose (e.g., law enforcement tasks, corporate investigations with legal mandate). Apply proportionality and necessity tests before prolonged tracking (see Section 5.5.3).

5.4.4. Handling unlawfully obtained data

- Definition: Downloading or storing data from breaches, darknet markets, or leaks may constitute unlawful possession or processing of stolen data, and its processing is legally borderline.

⁹ Due to the complex nature of all the relevant legislatures, it is impossible to present here a sufficiently brief and yet exhaustive legal analysis that would serve as a universal defense. The “prevention” recommendations are therefore strictly that — recommendations. Nothing more.

¹⁰ This would also potentially trigger general limitations posed on AI systems in connection to the synthetically generated content and many restrictions falling under the Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonized rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act).

- Prevention: If breach data must be analyzed (e.g., for defensive cybersecurity purposes), avoid storing full datasets, work with metadata or hashes, and ensure clear documentation of purpose and legal basis. When possible, rely on official breach-notification services rather than raw leak archives. When the situation demands analyzing full datasets and the risk cannot be mitigated otherwise, try to use minimum of the data.

The Budapest Convention provides a harmonized international legal framework for most of these offenses and international cooperation in investigation (including electronic evidence collection). For OSINT practitioners, this means a certain level of transparency. If they are located in a signatory state and a certain action is unlawful there, according to a national criminal law, they should be wary of taking this action in any of the signatory states.

Besides harmonizing the definitions of cybercrimes, the Budapest Convention also provides guidance on the classification and handling of electronic evidence by law enforcement agencies, establishing standards for admissibility in judicial proceedings. The Budapest Convention establishes standardized investigative powers including expedited preservation of computer data, production orders, search and seizure powers, and real-time data collection. These powers are essential for OSINT activities that involve electronic evidence. In Articles 23–35 the Budapest Convention establishes mutual legal assistance frameworks enabling cross-border OSINT investigations, also mentioning that it is required to cooperate “to the widest extent possible” in cybercrime investigations between relevant participants.

5.5. General problematic aspects and legal challenges

From a practical perspective, OSINT investigators – whether journalists, private companies, or law enforcement – face different but often overlapping challenges. In this section, we identify problematic areas in connection with the presented legislation and provide contextual recommendations for individual scenarios as introduced in Section 3, indicating which legislation is relevant to them in a summary table and, in the case of a given scenario, which legislation must be followed.

Table 10 presents a detailed legal compliance assessment for the provided scenarios. It is seen that some of the activities presented in the scenarios are on the edge of, or even beyond, what is permitted by legal regulations. However, the activities included in most scenarios are often feasible (after taking the appropriate steps), and the legal framework does not fundamentally limit the use of OSINT tools.

5.5.1. Jurisdictional complexities

Law enforcement OSINT activities frequently involve cross-border data flows and multi-jurisdictional investigations, creating complex legal compliance challenges. The extraterritorial application of GDPR to any processing of EU citizens’ data regardless of processor location compounds these difficulties. Cloud-based investigations exemplify these challenges, where data may be processed across multiple jurisdictions with varying legal requirements. In cases where data are nevertheless released, an accelerated procedure for its provision and exchange has been introduced through the EPOC Regulation (European Union, 2023).

5.5.2. Consent validity and practical implementation

While consent appears straightforward in many of the introduced scenarios, obtaining valid GDPR-compliant consent for OSINT activities presents practical challenges. Consent must be freely given, specific, informed, and unambiguous, requirements difficult to satisfy in investigative contexts where the full scope of data processing may be unknown at the outset.

5.5.3. Legitimate interests assessment complexity

The majority of OSINT scenarios rely on the legitimate interests legal basis, requiring complex three-part tests: (i) legitimate interest iden-

Table 10

Legal compliance assessment for OSINT scenarios.

Scenario	Scenario title	Compliance assessment	Primary issues	Key regulations
1	Data breach investigation on darknet	Mixed compliance	Self-data search compliant; third-party search without consent and systematic storage non-compliant; other activities conditionally compliant	GDPR Art. 5, 6, 15; Data minimization principles, criminal law (Budapest Convention)
2	Online activity monitoring of children	Conditionally compliant	Balance between parental authority and child's evolving privacy rights; age-dependent considerations required	GDPR Art. 8; EU Charter Art. 7; UNCRC
3	Web application scanning for de-anonymization	Conditionally compliant	Legitimate interest for crime prevention; risk of processing third-party data; cross-border complications in cloud	GDPR Art. 6(1)(f); LED Budapest Convention
4	Person verification	Conditionally compliant	Legitimate interest assessment required; balancing test between verification needs and privacy rights	GDPR Art. 6(1)(f), Platform Terms & Conditions
5	Information gathering for legal proceedings	Conditionally compliant	Law enforcement exemptions may apply for criminal cases; legal proceedings as legitimate interest for civil cases	LED; GDPR Art. 6(1)(f), Art. 23
6	Identity disclosure	Conditionally compliant	Judicial proceedings exemption available; threat prevention as legitimate interest requires proportionality	GDPR Art. 6(1)(f), Art. 23; LED
7	Person search using open sources	Mixed compliance	Missing person search compliant (vital interests); other searches conditionally compliant based on legitimate interests	GDPR Art. 6(1)(d), 6(1)(f); LED; Budapest Convention
8	Competitive intelligence	Conditionally compliant	Commercial legitimate interest vs. privacy rights; proportionality assessment required for all activities	GDPR Art. 6(1)(f), Platform Terms & Conditions
9	Open source information gathering	Mixed compliance	Activities without personal data compliant; personal data processing requires legal basis; household exemption may apply	GDPR Art. 2(2)(c), Art. 6(1)(f), Platform Terms & Conditions, Database Protection
10	Creating fake profiles for OSINT	Mixed compliance	Passive profiles conditionally compliant; active profiles and real photo use non-compliant; violation of Terms & Conditions	Criminal law; Platform Terms & Conditions; GDPR Art. 6; AI Act

tification, (ii) necessity assessment, and (iii) balancing against data subjects' rights. This assessment varies significantly across contexts, from competitive intelligence to personal safety verification (Scenario 4).

5.5.4. Child protection and parental authority balance

Scenario involving children (Scenario 2) presents particular complexity in balancing parental authority with children's evolving privacy rights. It requires age-appropriate consideration of children's ability to consent to data processing, creating uncertainty for OSINT practitioners.

5.5.5. Platform deception and identity fraud

False profile creation (Scenario 10) raises multiple legal issues beyond data protection, including platform terms of service violations, identity fraud, and criminal impersonation (EITHOS, 2024). The use of AI-generated images (Scenario 10) introduces additional considerations under emerging AI regulation.

5.5.6. Data breach exploitation ethics

Processing data from known breaches (Scenario 1) raises fundamental questions about the legitimacy of exploiting unlawfully obtained data, even when the OSINT practitioner did not cause the breach. The systematic storage and processing of data from breaches (Scenario 1) is particularly problematic under data protection principles.

5.6. Summary

Legal risks in OSINT do not stem from the use of advanced tools alone but from the context of data acquisition and reuse. The public availability of information does not eliminate the rights of data subjects, content creators, or platform operators. Consequently, OSINT practitioners must remain aware that their activities may result in fines, civil suits, criminal charges, or loss of access to data sources. Therefore, it is highly recommended that they carefully assess the legal basis for each processing activity; avoid systematic scraping of

platforms without considering personal data protection, IP or Terms of Service implications; treat data from breaches or darknet sources as highly problematic; document legitimate interest assessments and data minimization measures and finally (and once again) distinguish between truly open data (open government data, open archives) and data that are merely visible online but legally restricted.

6. Conclusion

Information represented in data has become a valuable commodity today. Consequently, OSINT has evolved into a critical tool across various sectors, driven by the exponential growth of data available on the internet. Its applications span intelligence, law enforcement, cybersecurity, corporate investigations, and more, enabling the analysis of publicly accessible information to uncover valuable insights.

Based on our findings, all open sources are associated with tools, services, and databases for data collection and analysis. Most of these solutions are available either as open-source or without requiring paid access. However, their capabilities vary greatly, and there is no universal solution for OSINT purposes. Therefore, combining multiple tools with manual techniques is necessary to achieve the most accurate and verified intelligence.

At the same time, OSINT's growing relevance raises complex legal and ethical challenges. The distinction between information that is merely visible online and information that is lawfully reusable is not always clear, creating risks for practitioners who operate across diverse legal jurisdictions. These challenges highlight that the future of OSINT will depend not only on technological innovation but also on developing robust legal awareness, compliance strategies, and ethical standards. Without this balance, the value of the intelligence gained may be undermined by legal consequences or reputational damage.

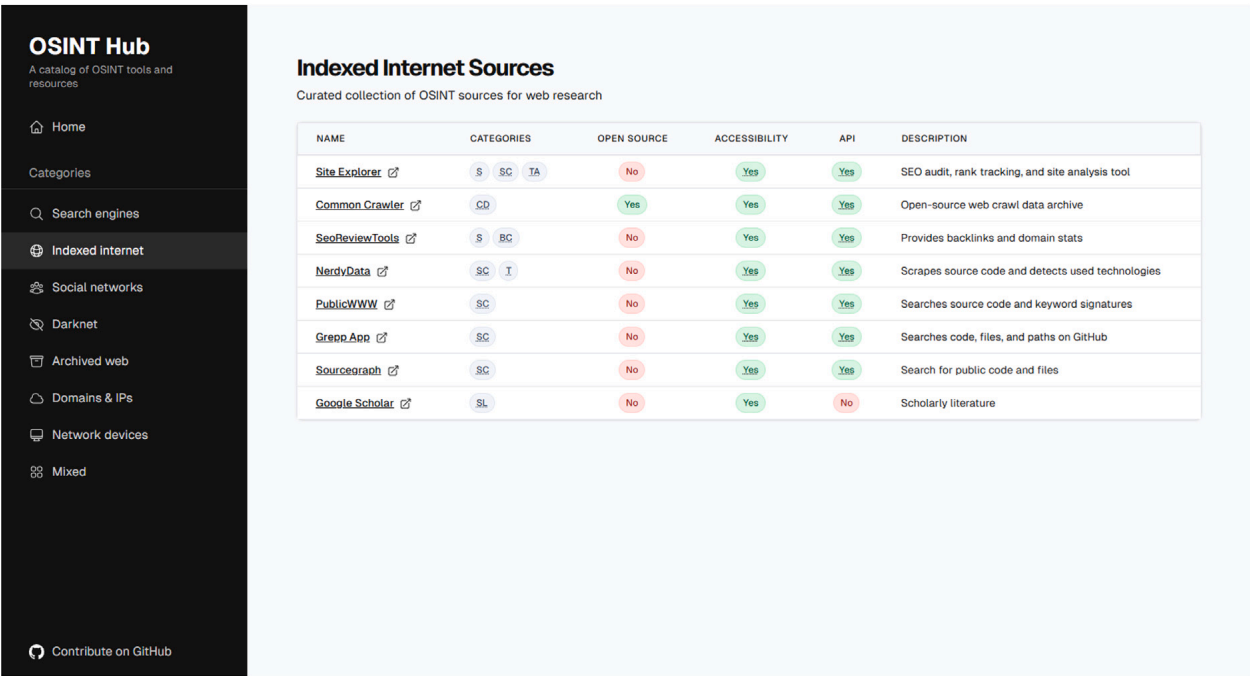


Fig. 2. Open website presentation of the OSINT comparative analysis.

CRediT authorship contribution statement

Willi Lazarov: Writing – original draft, Validation, Resources, Methodology, Investigation, Conceptualization, Data curation. **Vojtech Moravec:** Writing – original draft, Investigation, Data curation. **Pavel Loutocky:** Writing – original draft, Validation, Investigation. **Jakub Vostoupal:** Writing – original draft, Investigation. **Zdenek Marti-nasek:** Supervision, Project administration, Funding acquisition.

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Declaration of competing interest

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Appendix. Study dataset

We have published the comparative analysis dataset in an open-source GitHub repository. The repository is available at <https://github.com/larescze/osint-hub>, with a versioned snapshot archived on Zenodo (Lazarov et al., 2025a). Following the repository instructions, the dataset can be freely maintained and extended by the community. In addition, we have developed an interactive web presentation that showcases all the compared solutions. An example of data presentation is shown in Fig. 2 and the website can be accessed at <https://larescze.github.io/osint-hub/>.

Data availability

The data have been published on GitHub (<https://github.com/larescze/osint-hub>) and are accessible on a public website (<https://larescze.github.io/osint-hub/>). The versioned dataset has been published on Zenodo (<https://doi.org/10.5281/zenodo.19236684>).

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Willi Lazarov is a researcher at the Faculty of Electrical Engineering and Communication, Brno University of Technology (BUT), Czech Republic, where he specializes primarily in research on profiling, interactivity, and feedback methods to improve the effectiveness of cybersecurity education. Within security research projects, he also focuses on ethical hacking, static analysis, and open source intelligence. He received his MSc. at BUT in 2024 and is currently completing his PhD. He also actively cooperates with the Czech Army, National Cyber and Information Security Agency, University of Defence, and research groups at in Finland and Norway. In recognition of his contributions, he has received awards such as the EY Cyber Security Trophy, EY ESO Cyber Security Future Promise, Rector's Award, and Minister of the Interior Award for outstanding achievements in security research.



Vojtech Moravec is an analyst at Gen Digital, specializing in malware analysis and detection engineering. Since 2022, he has participated in several research projects on penetration testing and open source intelligence at the Faculty of Electrical Engineering and Communication, Brno University of Technology (BUT), Czech Republic. He received his bachelor's degree from BUT in 2024.



Pavel Loutocký is an assistant professor and post-doctorate researcher at the Institute of Law and Technology, Faculty of Law, Masaryk University and Faculty of Informatics, Masaryk University. He participated in a number of national and international research projects and delivered lectures concerning online dispute resolution, electronic identification and documents, online consumer protection, cybersecurity or online platforms. Since 2012, he has been a regular observer of selected working groups within the United Nations Commission on International Trade Law (UNCITRAL). He is also an external lecturer at the Georg-August-Universität Göttingen, Danube University Krems or University of Pécs and became a fellow of the National Center for Technology & Dispute Resolution.



Jakub Vostoupal is an assistant professor at the Department of International and European Law, Faculty of Law, Masaryk University. His research focuses primarily on cybersecurity law, including topics such as the legal aspects of ethical hacking, cybersecurity certification, and the application of international law in cyberspace. He has taken part in numerous national and international research projects addressing diverse dimensions of cybersecurity. In addition to his academic work, he serves as the cybersecurity manager at both the Czech Supreme Court and the Czech Supreme Administrative Court. He is also pursuing studies in psychology, with a particular interest in the intersection of law, technology, and human behavior.



Zdenek Martinasek is an associate professor at the Faculty of Electrical Engineering and Communication at Brno University of Technology (BUT), Czech Republic. He completed his M.Sc. degree in 2008 and received his Ph.D. degree in 2013. His research activities focus on applied cryptography, side-channel cryptanalysis, and cybersecurity. His biggest expertise is in power analysis attacks based on machine learning. Currently, he is involved as a task leader in security research projects of the Ministry of Interior of the Czech Republic. He has published more than 45 papers in international journals and conferences and has provided several invited research and teaching lectures focused on cybersecurity. During his Ph.D. studies, he completed an internship at TU Wien. He also helps to cover pedagogically Bachelor's and Master's program courses in Information Security.